Algorithms

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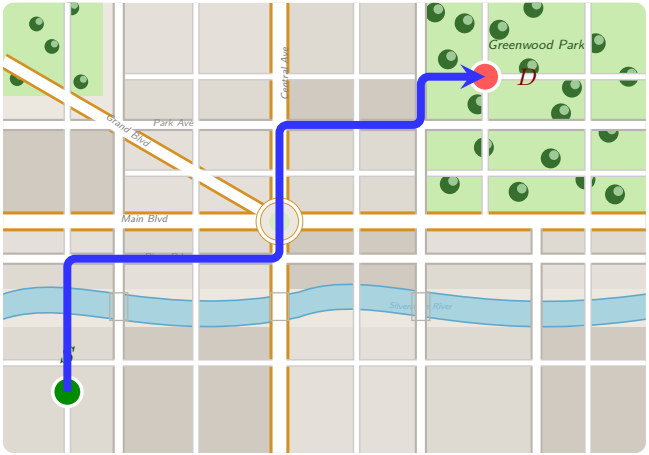
Motivation: Routing at Scale



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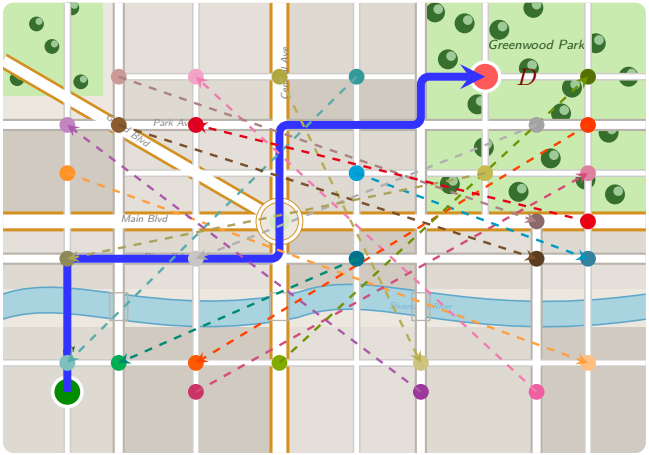

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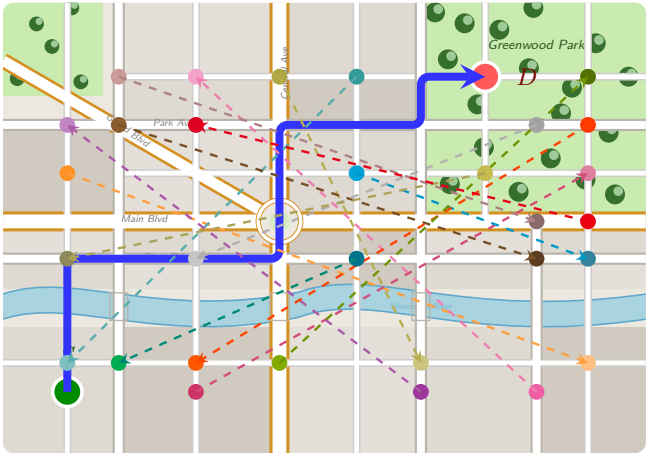

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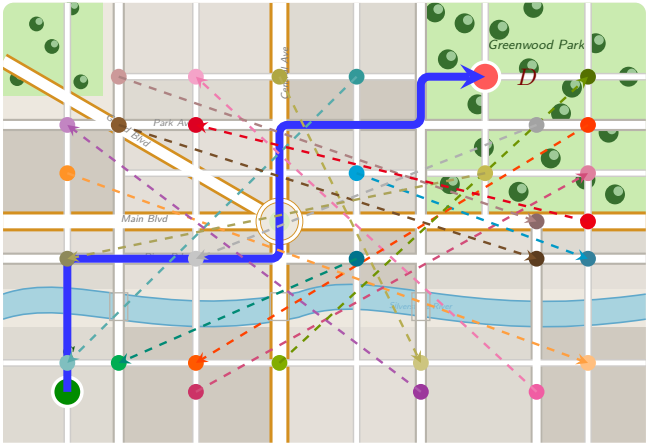

Idea: Precompute all shortest paths (APSP).

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All-Pairs Shortest Paths

Given a weighted graph $G = (V, E, w)$, compute shortest-path distances between **every ordered pair** (u, v).

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Use cases

Distance queries in road networks, routing between all servers, and optimization pipelines.

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Index vertices as $1, 2, \ldots, n$.

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After $k = n$, matrix $D^{(n)}$ is the APSP answer.

Floyd–Warshall Pseudocode

```
procedure FloydWarshall(W)
    D ← W    (adjacency matrix, with ∞ for missing edges)
    for i = 1 to n do: D[i][i] ← 0
    for k = 1 to n do
      for i = 1 to n do
        for j = 1 to n do
          if D[i][k] + D[k][j] < D[i][j] then
            D[i][j] ← D[i][k] + D[k][j]
    return D
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Complexity: triple loop $\Rightarrow O(V^3)$ time, $O(V^2)$ space.

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Practical output convention

Report pair (i, j) as $-\infty$ if there exists a vertex k with $D[k][k] < 0$, reachable from i and reaching j.